# J.VENKATESH

Electronics Engineering Student | Aspiring Web Development Engineer

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**LINKED IN** https://www.linkedin.com/in/venkatesh-j-3b293628a | **GITHUB** VenkateshJ23BEC1231/Project-Files-and-Reports
https://github.com/23f2001690


## PROFESSIONAL SUMMARY

Electronics Engineering student with hands-on project experience in Verilog-based RTL design and web application development. Familiar with open-source RTL-to-GDSII flow (Qflow, Magic VLSI) and basic physical design concepts.Would also like to further work on web development projects.

## EDUCATION

**Vellore Institute of Technology (Chennai)** Aug 2023 - Jul 2027
*Bachelor of Technology - Electronics and Communication Engineering* **Current CGPA: 8.72/10**

**BS Diploma (Ongoing)** **CGPA:7.98 Sep 2024-present**
*Indian Institute of Technology, Madras*
*Data Science and Programming*

**BS FOUNDATIONAL LEVEL: COMPLETED**

**Relevant Coursework:** Digital Electronics, Microcontrollers, VLSI Design, Signal Processing, Embedded Systems,Advanced Cloud Computing(AWS) along with IITM Foundation and Diploma courses.

## TECHNICAL SKILLS

**Programming Languages:**
Verilog HDL, VHDL,Python, Java, C and Embedded C,HTML,CSS,Javascript

**EDA and CAD Tools:**
Cadence Virtuoso (Basic),Quartus and ModelSim, Yosys, Netgen, Magic (DRC), Qrouter, Graywolf, Qflow, KiCAD

**Core Areas:**
Digital Logic Design, RTL Design (Academic Projects), Basic Physical Design Exposure, Data Science, Web Development with Python

**Data Related Tools**: LabVIEW,RStudio,MATLAB,Keil

## PROFESSIONAL EXPERIENCE

### Physical Design Implementation of SPI Protocol | MAY-JULY 2025| TECHNICAL INTERNSHIP

Technologies: Qflow, Yosys, Graywolf, Magic, OpenSTA, Qrouter, Netgen, Verilog RTL
- Completed end-to-end physical design implementation of SPI protocol using Qflow toolchain
- Performed RTL synthesis with Yosys converting Verilog to gate-level netlist
- Executed placement and routing using Graywolf and Qrouter for optimized signal delays
- Conducted Static Timing Analysis using OpenSTA to verify timing requirements
- Generated final GDSII file for chip fabrication with DRC and LVS verification

### AI with Python Internship Program |APR-MAY 2023

Technologies: Python, TensorFlow, Scikit-learn, NumPy, Pandas, Jupyter
Completed intensive 2-month AI and Machine Learning training program in collaboration with Microsoft
Implemented supervised and unsupervised learning algorithms including decision trees, neural networks, and clustering techniques
Developed end-to-end machine learning pipeline covering data preprocessing, feature engineering, model training, and evaluation
Built practical AI applications using Python libraries and frameworks for real-world problem solving

## KEY PROJECTS

### ML-Integrated Ukulele Chord Detection System

Technologies: Python, Arduino UNO, Machine Learning, Mobile App Dev, Signal Processing
- Developed innovative dual-sensing chord detection system combining hardware sensors with audio analysis
- Designed Arduino-based hardware circuit using Force Sensitive Resistors for finger placement detection
- Created mobile application interface with real-time audio recording and ML-based chord classification
- Implemented voltage divider circuits with 10kΩ resistors for accurate pressure detection
- Designed accessible learning tool specifically beneficial for visually impaired users and beginners

### Statistical Analysis of Government Healthcare Subsidies

Technologies: R, RStudio, ggplot2, Excel, Statistical Modeling, Hypothesis Testing
- Conducted comprehensive analysis of medical subsidies effectiveness across India, Australia, Germany, and USA
- Designed and executed primary data collection survey gathering 50+ responses from VIT Chennai students
- Performed paired T-test hypothesis testing at 5% significance level to analyze subsidy impact on medical expenditure
- Implemented linear regression modeling to predict future Out-Of-Pocket Expenditure (OOPE) trends
- Developed predictive model showing India reaching sub-10% OOPE by 2049 with current trend of -1.47% annual reduction

### Sleep Monitoring System using LabVIEW

Technologies: LabVIEW, Hardware Sensors, Data Acquisition, CSV Processing, Machine Learning
- Developed comprehensive sleep monitoring system using LabVIEW for real-time physiological parameter tracking
- Integrated multiple hardware sensors including SpO2, room temperature, humidity, body temperature, and heart rate monitors
- Implemented data acquisition pipeline collecting sensor parameters and generating CSV files for ML processing

- Created LabVIEW block diagram with real-time data visualization and automated file handling capabilities

### Trekking Management web-Application using Flask

Technologies: Flask, Jinja2, Bootsrap CSS, JS,SQLite

- Built a full-stack Flask trekking management app with role-based access control for Admin, Staff, and User roles.
- Designed a normalized SQLAlchemy database schema with four related models and proper foreign-key relationships.
- Implemented CRUD workflows for trek management, bookings, and staff approvals with real-time slot tracking.
- Built secure session-based authentication with a custom reusable login/role-check decorator.
- Developed a responsive Bootstrap and Jinja2 frontend with role-specific dashboards and search/filter functionality.

**Placement Portal Application (PPA) — Full-Stack Web App using Flask & Vue.js**

Technologies: Flask, Flask-SQLAlchemy, Flask-JWT-Extended, Vue.js 3, Bootstrap CSS, Celery, Redis, SQLite

- Built a full-stack campus placement management platform with JWT-based role access control for Admin, Company, and Student roles, backed by a unified user model.
- Designed a normalized SQLAlchemy schema with six related models (User, Student, Company, JobPosition, Application, Placement), enforcing referential integrity via foreign keys and cascading deletes.
- Implemented end-to-end placement workflows — company registration and admin approval, drive posting, student applications with server-side eligibility validation (CGPA/branch/year), and status tracking (like applied, placed etc.,).
- Engineered asynchronous background jobs with Celery and Redis, including scheduled daily interview reminders, monthly HTML placement reports emailed to companies, and user-triggered CSV export of application history with real-time polling.
- Added Redis-based API response caching and event-driven cache invalidation to optimize high-traffic admin and student endpoints.
- Developed a responsive Vue.js 3 frontend (CDN-based, no build tooling) with role-specific dashboards, reusable modal components, and dynamic search/filter for companies, students, and drives.
- Enforced application integrity with database-level unique constraints to prevent duplicate applications, alongside blacklist/deactivation controls for admin-managed access revocation.

## CERTIFICATIONS

*MAVEN SILICON PHYSICAL INTERNSHIP CERTIFICATE
*BS FOUNDATION CERTIFICATE
*NPTEL-VHDL BASED SIMULATION AND SYNTHESIS OF DIGITAL CIRCUITS (WORSKSHOP COMPLETION CERTIFICATE)
*NPTEL- WILDLIFE ECOLOGY AND FORESTS AND THEIR MANAGEMENT (ELITE CERTIFICATES)